

Lab Practical #13:

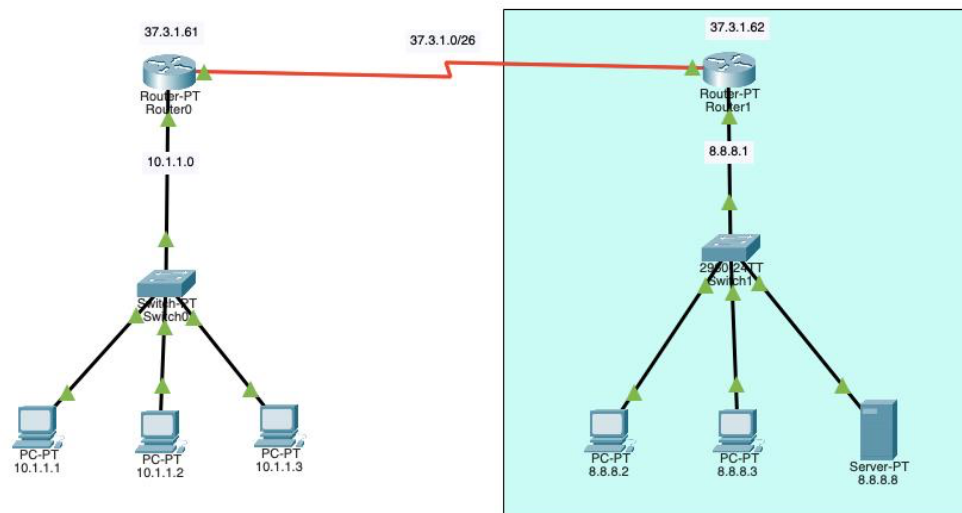
Study and Configure Network Address Translation.

Practical Assignment #13:

A. To configure Static NAT and Port Address Translation (PAT) on a Cisco Router and verify communication between private and public networks.

To do Instructions:

1. Required followed things: PDU details, NAT TABLE, Topology, Static routing table, Router CLI (performed commands)



👉 NETWORK ADDRESS TRANSLATION STEP BY STEP:

1. Open Cisco Packet Tracer

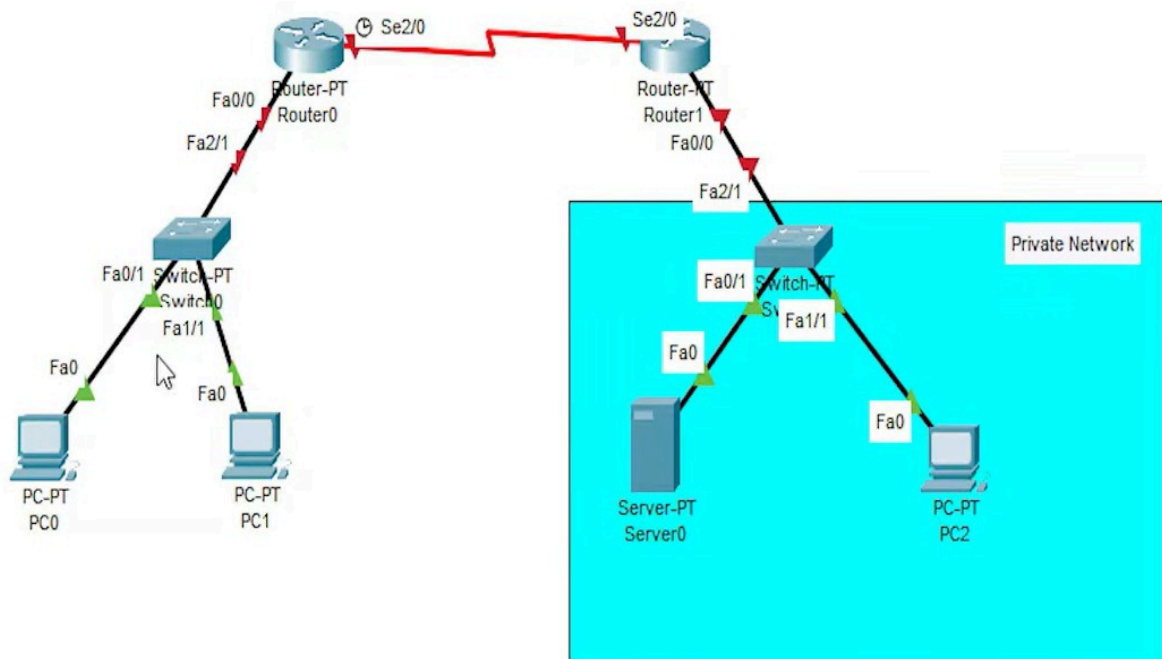
Open **Cisco Packet Tracer** and create a new project.

2. Add the devices

Add these devices:

- **2 Router** — 2911 OR PT-ROUTER
- **2 Switch** — 2960 OR PT-SWITCH
- **3 PC** — PC0 ,PC1,PC2
- **1 Server** — Server0

Your topology will look like:



3. Connect the devices

Use **Copper Straight-Through Cable**.



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4. Configure Each PC0-PC1-PC2 & SERVER

Click **PC0** → **Desktop** → **IP Configuration**.

>> PUBLIC NETWORK:

PC0

IP Address: 10.0.0.2

Subnet Mask: 255.0.0.0

Default Gateway: 10.0.0.1

PC1

IP Address: 10.0.0.3

Subnet Mask: 255.0.0.0

Default Gateway: 10.0.0.1

>> PRIVATE NETWORK

PC2

IP Address: 20.0.0.2

Subnet Mask: 255.0.0.0

Default Gateway: 20.0.0.1

SERVER

Click **Server0** → **Desktop** → **IP Configuration**.

IP Address: 20.0.0.3

Subnet Mask: 255.0.0.0

Default Gateway: 20.0.0.1



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PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

10.0.0.2

Subnet Mask

255.0.0.0

Default Gateway

10.0.0.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

 /

Link Local Address

FE80::20B:BEFF:FE75:D739

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

GIVE IP ADDRESS LIKE THIS IN EACH DEVICES:-----

5. Configure Router0 (Public Network Router)

Router0 connects the public network (10.0.0.0/8, PC0 & PC1) to Router1 through a serial WAN link. Click Router0 → Config tab and set the interfaces as below.

FastEthernet0/0 (LAN side – connects to Switch0 / PC0, PC1)

Port Status : On

IPv4 Address : 10.0.0.1

Subnet Mask : 255.0.0.0

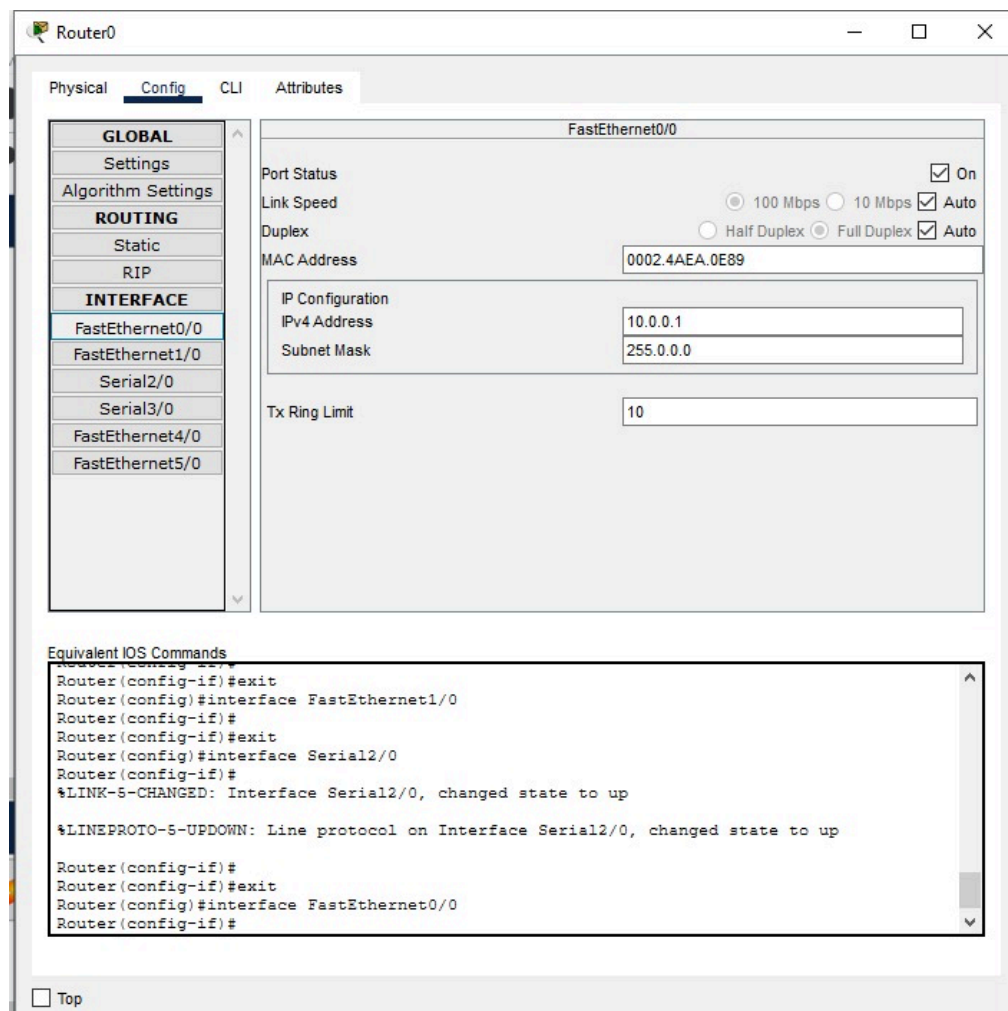


Fig 5.1 – Router0 FastEthernet0/0 Configuration

Serial2/0 (WAN side – connects to Router1)

Port Status : On

Clock Rate : 2000000 (Router0 is the DCE end of the link)

IPv4 Address : 30.0.0.1

Subnet Mask : 255.0.0.0

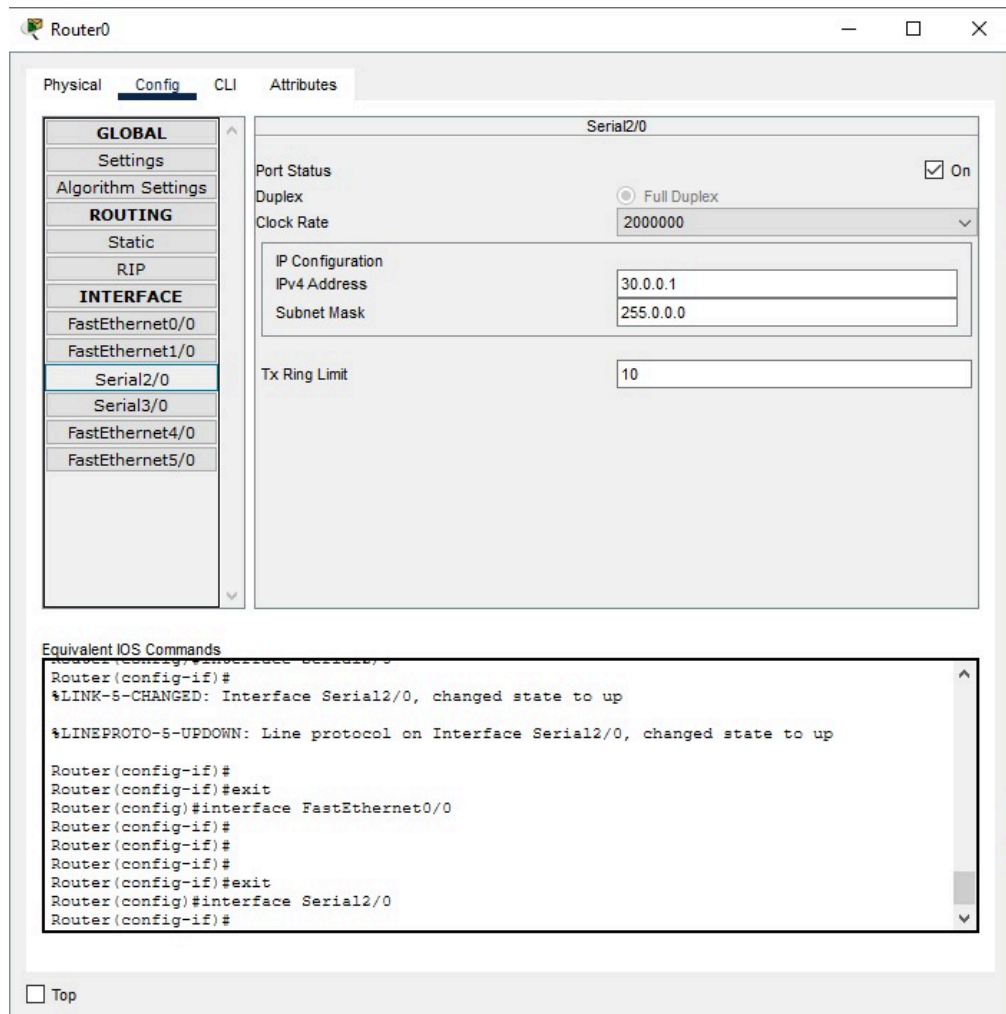


Fig 5.2 – Router0 Serial2/0 Configuration

6. Configure Router1 (Private Network / NAT Router)

Router1 sits between the private network (20.0.0.0/8, PC2 & Server0) and the serial link to Router0. This is the router on which NAT / PAT will be configured.

FastEthernet0/0 (LAN side – connects to Switch1 / PC2, Server0)

Port Status : On

IPv4 Address : 20.0.0.1

Subnet Mask : 255.0.0.0

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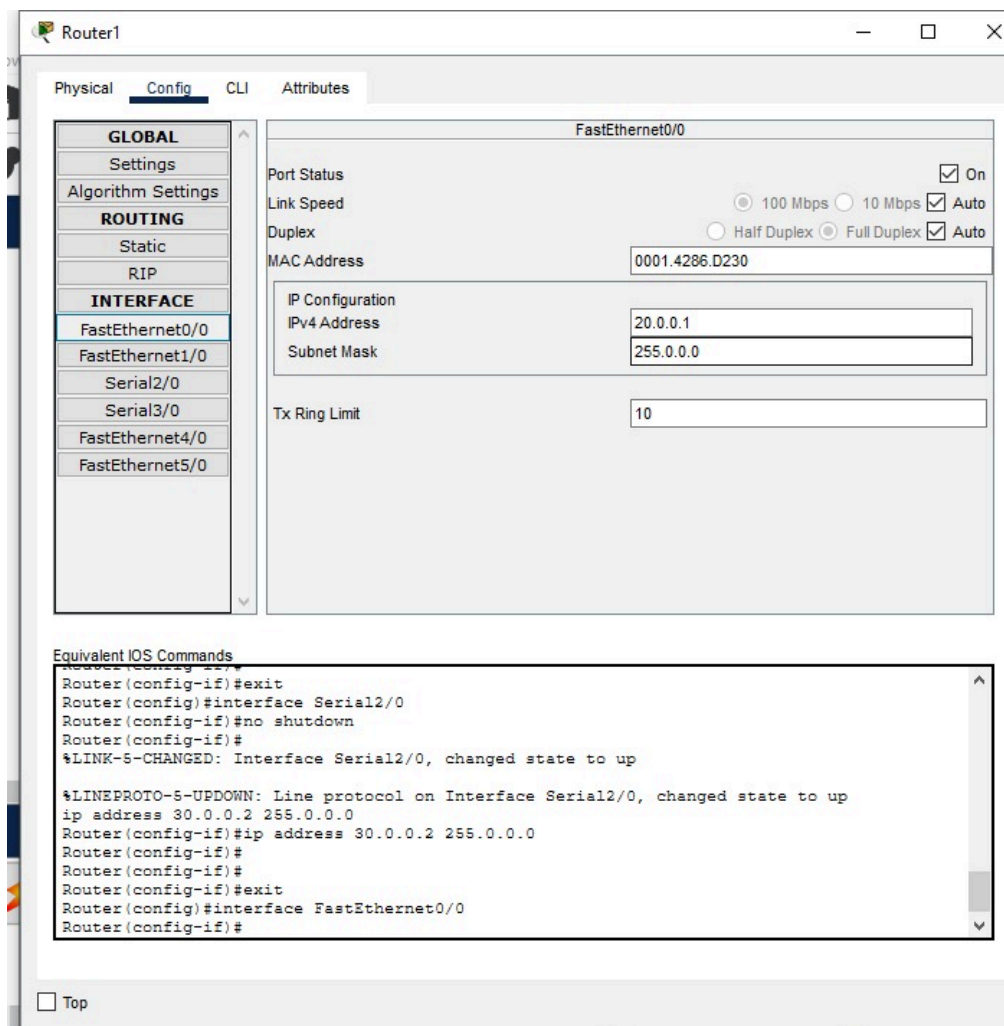


Fig 6.1 – Router1 FastEthernet0/0 Configuration

Serial2/0 (WAN side – connects to Router0)

Port Status : On

IPv4 Address : 30.0.0.2

Subnet Mask : 255.0.0.0

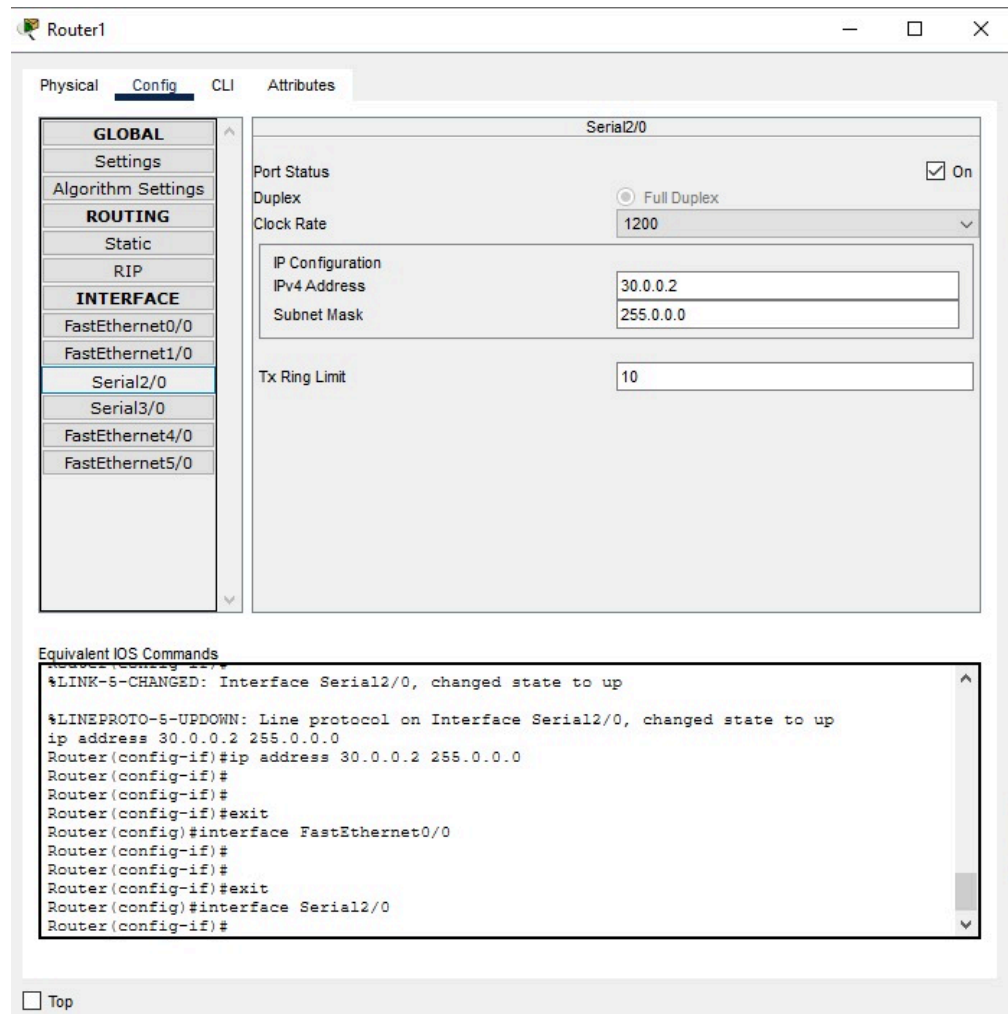


Fig 6.2 – Router1 Serial2/0 Configuration

7. Bring Up the Interfaces & Assign Addresses via CLI

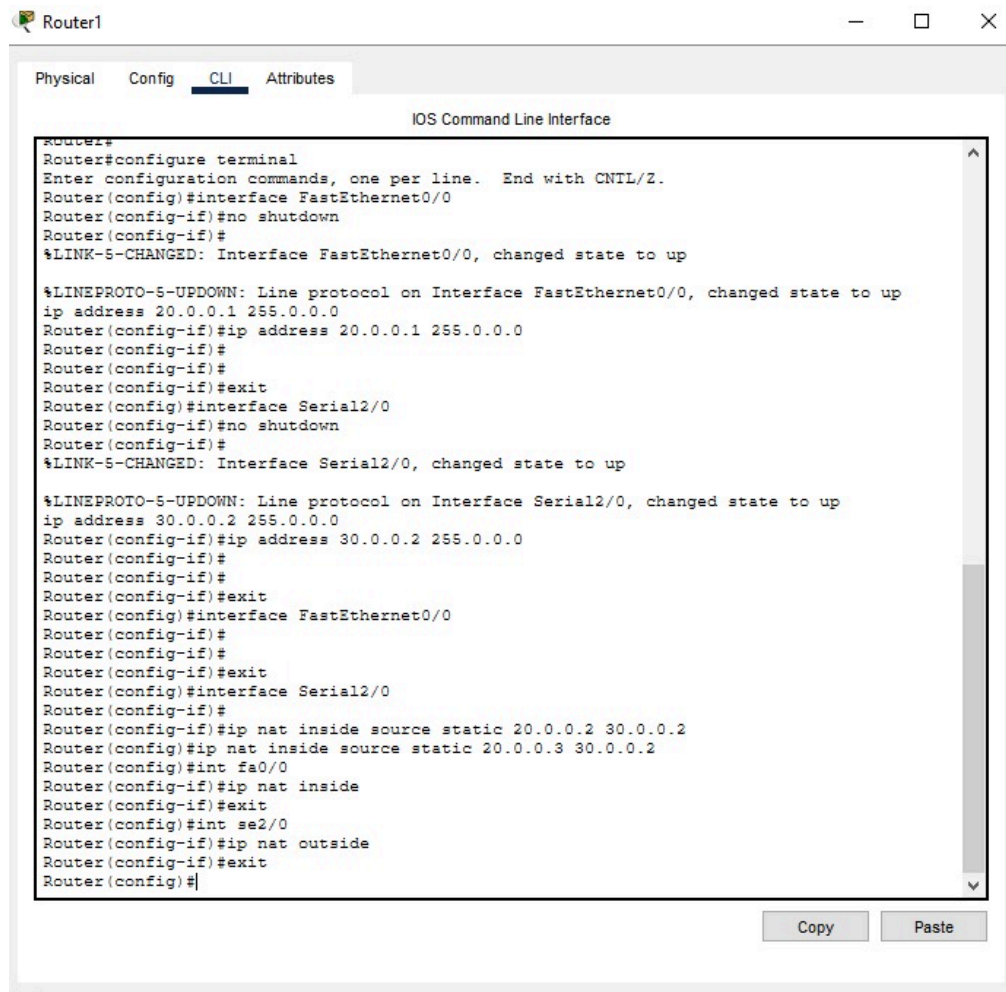
Open Router1 → CLI tab and enter global configuration mode, then configure and enable each interface:

```

Router>enable
Router#configure terminal
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#no shutdown
Router(config-if)#ip address 30.0.0.2 255.0.0.0
Router(config-if)#exit
  
```

The same procedure (no shutdown + ip address ...) is repeated on Router0 for FastEthernet0/0 and Serial2/0 (with the clock rate set, since Router0 is the DCE end).

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```
Router1
Physical Config CLI Attributes
IOS Command Line Interface
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 20.0.0.1 255.0.0.0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
ip address 30.0.0.2 255.0.0.0
Router(config-if)#ip address 30.0.0.2 255.0.0.0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#
Router(config-if)#ip nat inside source static 20.0.0.2 30.0.0.2
Router(config)#ip nat inside source static 20.0.0.3 30.0.0.2
Router(config)#int fa0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#int se2/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#
```

Fig 7.1 – Router1 CLI: interfaces enabled, addressed and NAT commands entered

8. Configure Static Routing on Router1

Router1 only knows about its own directly connected networks (20.0.0.0/8 and 30.0.0.0/8). To let return traffic reach the public 10.0.0.0/8 network, a static route pointing to Router0's serial interface is added on Router1 (Config → ROUTING → Static):

Network : 10.0.0.0

Mask : 255.0.0.0

Next Hop : 30.0.0.1

Equivalent CLI command:

```
Router(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
```

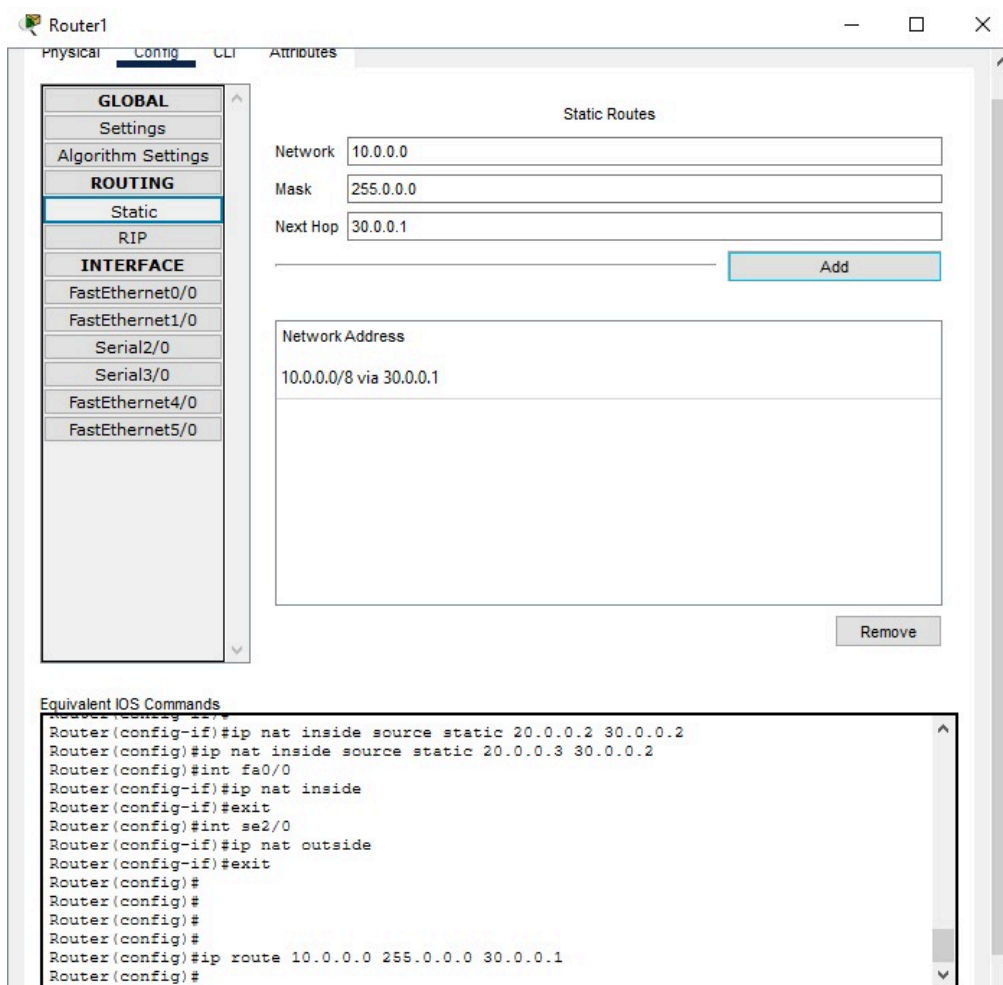


Fig 8.1 – Static Route (10.0.0.0/8 via 30.0.0.1) configured on Router1

9. Configure NAT (Static NAT) and PAT on Router1

Network Address Translation (NAT) lets the private 20.0.0.0/8 hosts communicate with the public 10.0.0.0/8 network using the router's public (global) address space on the 30.0.0.0/8 link.

Static NAT – creates a fixed, permanent one-to-one mapping between one inside-local (private) address and one inside-global (public) address. It is normally used for servers that must always be reachable at the same public address.

PAT (Port Address Translation / NAT Overload) – lets many inside-local addresses share a single inside-global address at the same time, by distinguishing each session using a different source port number. This is the form of NAT used when an organisation has more private hosts than public IP addresses.

Commands entered on Router1 (Config → CLI):

```
Router(config)#ip nat inside source static 20.0.0.2 30.0.0.2
Router(config)#ip nat inside source static 20.0.0.3 30.0.0.2
Router(config)#interface FastEthernet0/0
Router(config-if)#ip nat inside
```

PAT / NAT Configuration and Verification Screenshots

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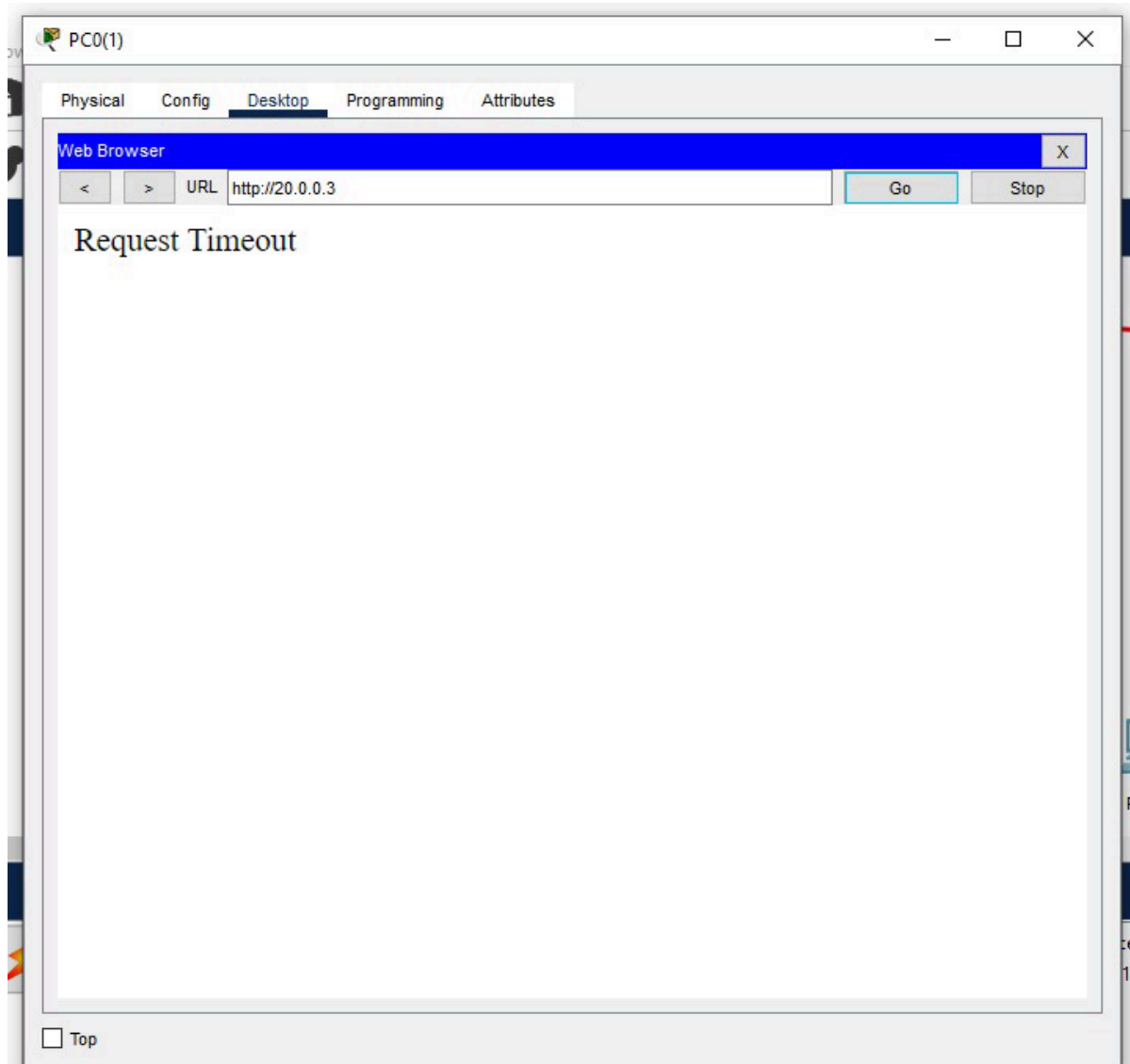


Figure 9.1 – PAT/NAT configuration or connectivity verification

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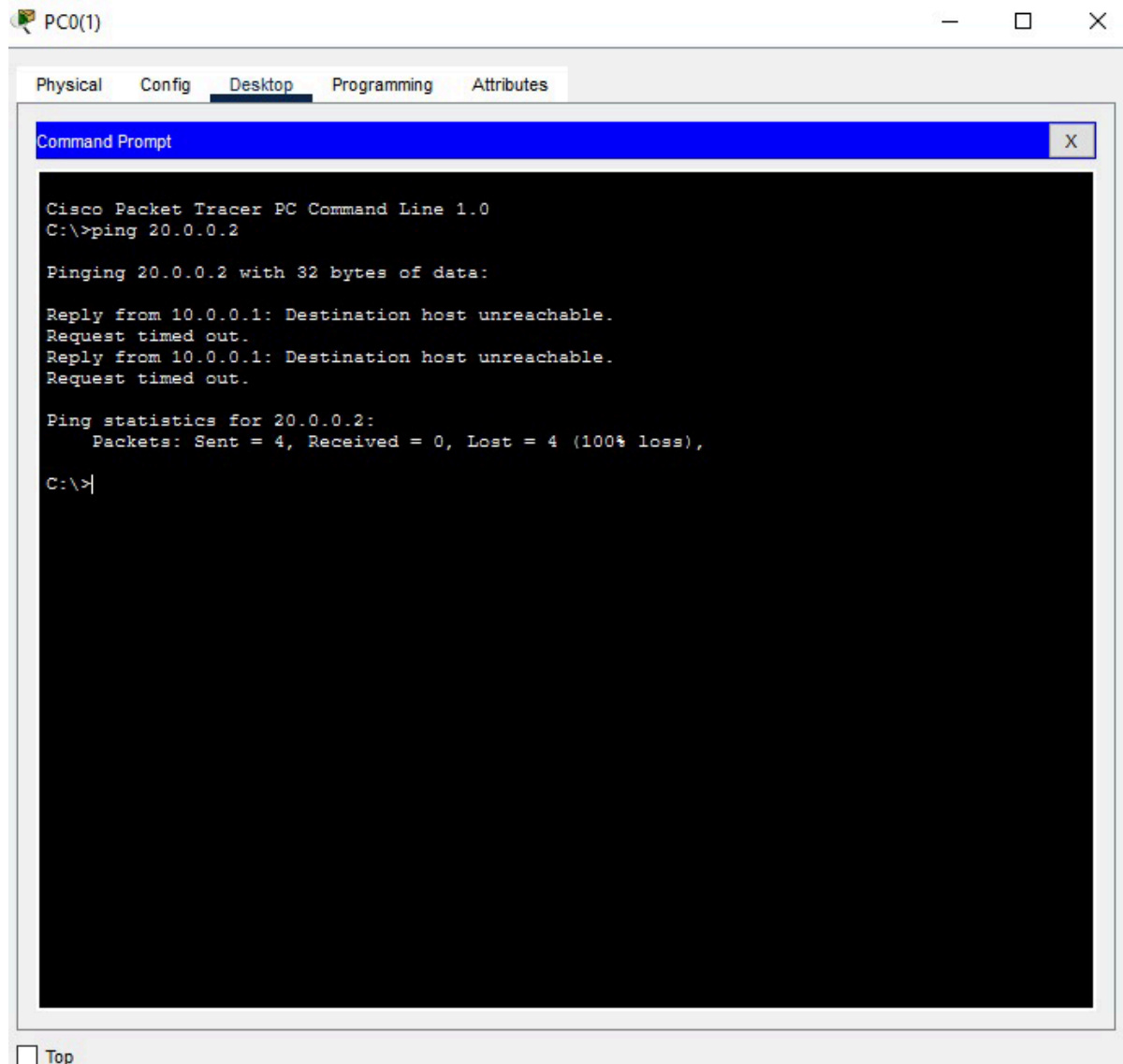
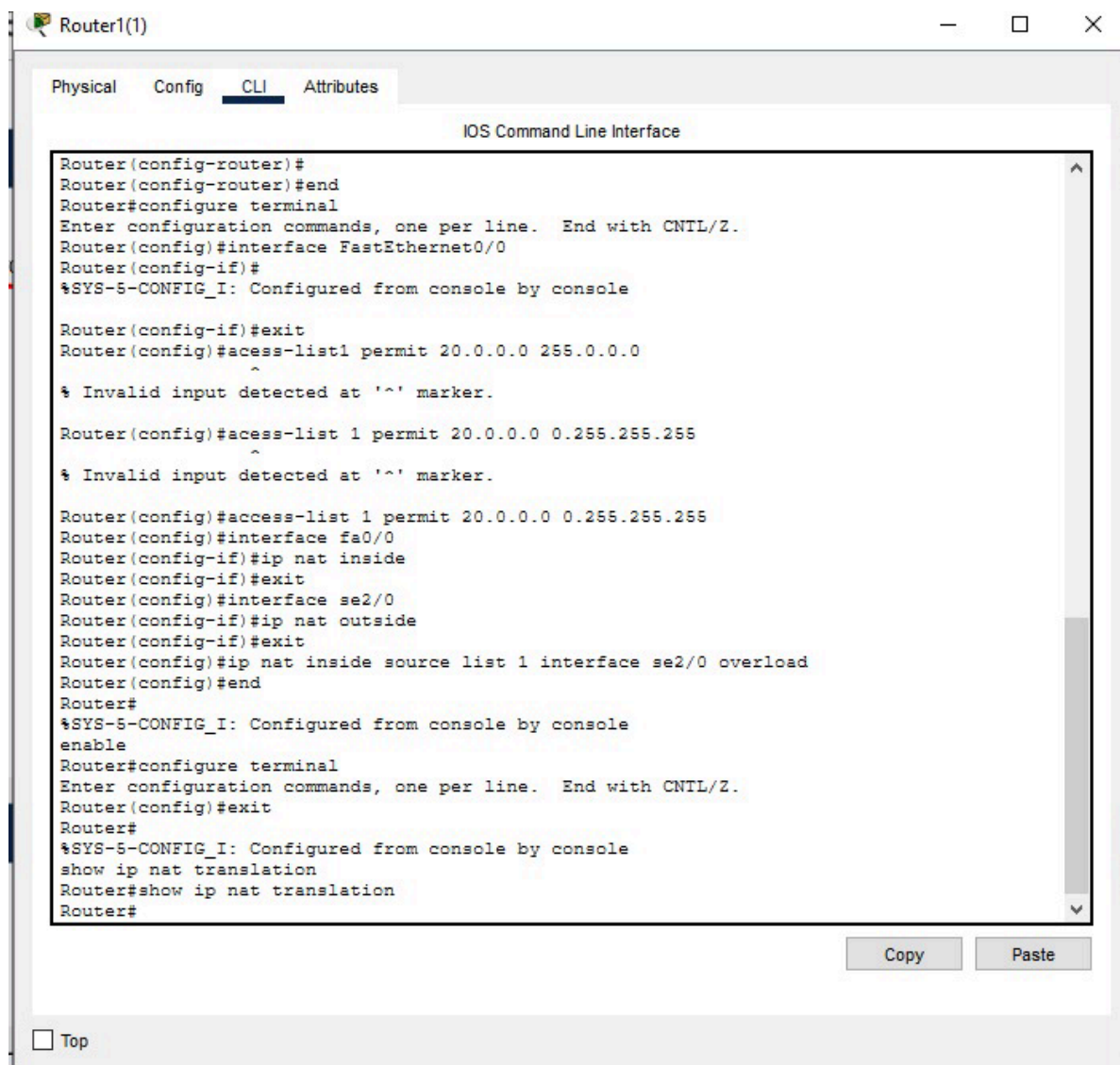


Figure 9.2 – PAT/NAT configuration or connectivity verification

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```

Router1(1)
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-router)#
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
%SYS-5-CONFIG_I: Configured from console by console

Router(config-if)#exit
Router(config)#access-list 1 permit 20.0.0.0 255.0.0.0
^
% Invalid input detected at '^' marker.

Router(config)#access-list 1 permit 20.0.0.0 0.255.255.255
^
% Invalid input detected at '^' marker.

Router(config)#access-list 1 permit 20.0.0.0 0.255.255.255
Router(config)#interface fa0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface se2/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip nat inside source list 1 interface se2/0 overload
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip nat translation
Router#show ip nat translation
Router#
  
```

Figure 9.3 – PAT/NAT configuration or connectivity verification

→ In Above Picture Cli Command For Pat Configuration:-

```

Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#ip nat outside
Router(config-if)#exit
  
```

Here fa0/0 (facing the private LAN) is marked ip nat inside, and se2/0 (facing Router0/the public network) is marked ip nat outside – this tells the router which side of each translated packet is “inside” and which is “outside”.

NAT Table (Inside Local → Inside Global)

Device	Inside Local (Private IP)	Inside Global (Translated / Public IP)
PC2	20.0.0.2	30.0.0.2

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Server0	20.0.0.3	30.0.0.2
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Note: In this practical both PC2 and Server0 were translated statically to the same global address 30.0.0.2. In real deployments each host normally needs its own unique inside-global address for plain static NAT; when several hosts must genuinely share one public address, the port-based static entries (e.g. ip nat inside source static tcp/udp ...) or a PAT/overload pool are used instead so that return traffic can be told apart by port number.

10. Verify NAT/PAT and Test Connectivity

On Router1's CLI, the active translations can be checked at any time with:

```
Router#show ip nat translations
Router#show ip nat statistics
```

From PC0 (public network, 10.0.0.2), a ping was sent to the private address of the server, 20.0.0.3:

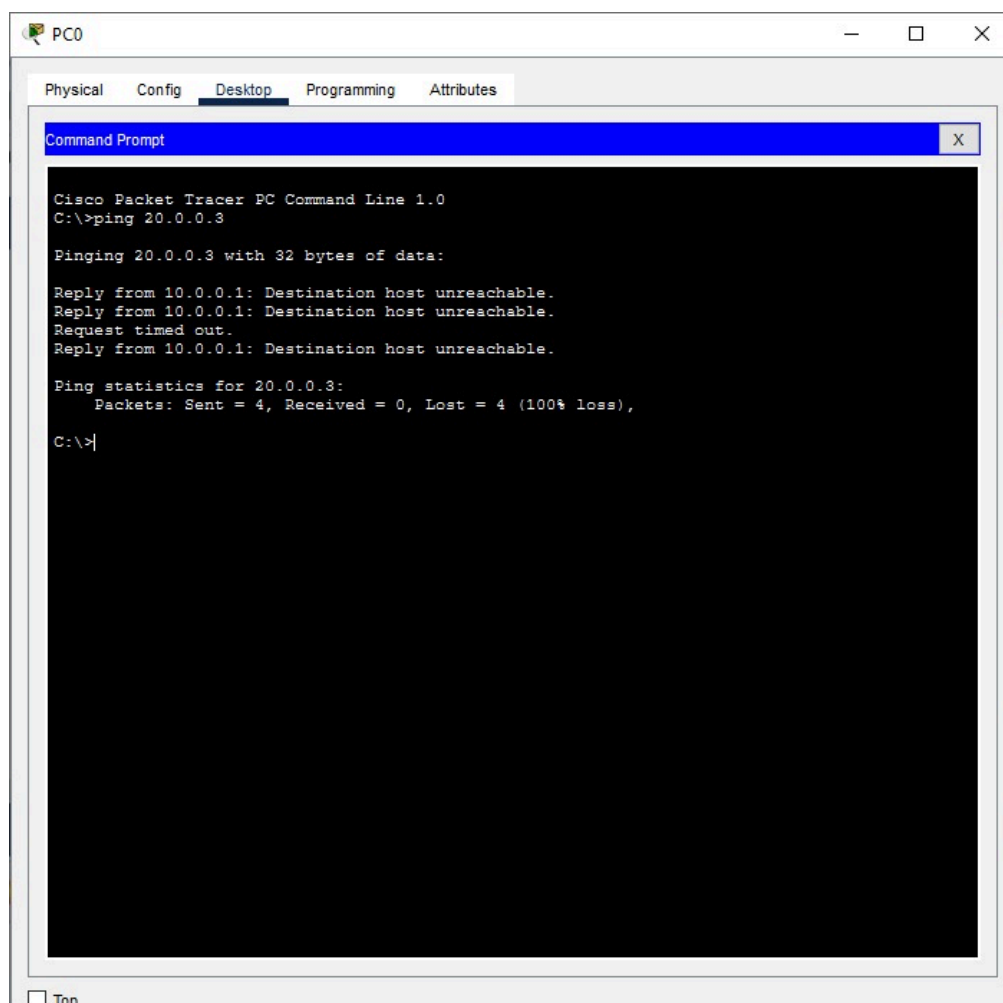


Fig 10.1 – Ping from PC0 to 20.0.0.3 – “Destination host unreachable”

Result and explanation: the ping to 20.0.0.3 fails, because 20.0.0.3 is an inside-local (private)



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address that is not advertised or routable outside the private network – Router0 has no route to the 20.0.0.0/8 network at all. To reach Server0/PC2 from the public side, the outside host must instead ping the translated inside-global address, 30.0.0.2 (e.g. ping 30.0.0.2 from PC0), which Router1 will translate back to the correct private host using the static NAT table above.

Conclusion

Router1 was successfully configured with Static NAT, translating the private hosts 20.0.0.2 (PC2) and 20.0.0.3 (Server0) to the public address 30.0.0.2 on its outside (Serial2/0) interface. A static route was added on Router1 so that reply traffic destined for the public 10.0.0.0/8 network is correctly forwarded back through Router0. This demonstrates how NAT/PAT allows hosts on a private, non-routable address space to communicate with a public network while keeping the internal addressing scheme hidden from the outside world.